

Novelty-Gated Compensation: What Causally Hijacking Actions Reveals About Self-Attribution in VLA Policies

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Abstract

Do learned robot policies internalize their own causal role in the world, or do they merely predict from what they see? We adjudicate with a causal audit: during closed-loop rollouts of 7B vision-language-action policies (OpenVLA-OFT) on four LIBERO suites, we covertly hijack the executed action chunk with probability 0.25 and ask whether anything inside the policy distinguishes self-caused from hijacked transitions. Across seven runs—a discovery run, a pre-registered seed replication, three further suites with their own checkpoints, and two gross-violation arms—no hidden-state probe, linear or one-hidden-layer MLP, separates from a selection-matched shuffled-label floor wherever the executed outcome is kinematically plausible. The null carries its own positive controls. Comparison features of command and outcome decode the same label at 0.69–0.91 in every run; a 256-unit MLP handed only the raw command $\oplus$ outcome concatenation *computes* the comparison (0.69 vs. 0.46 for its linear counterpart)—swap the outcome half for hidden states and decoding collapses to floor. The ingredients are present (the commanded plan reads out of the pre-action state at $R^2=0.64$, replicating at the same layer and pool), and the computation is demonstrably cheap; the policy’s states never expose it. The policy does react behaviorally—but the reaction is gated by the *visual novelty* of the outcome, not by the decodable command–outcome mismatch: significant after visibly reversed motion ($p \approx 0.01$, $n=119$ episodes), absent after the mechanically most blatant intervention, freezing ($n=115$), and small and seed-fragile for stealthy swaps (significant in 2 of 5 arms; null in the pre-registered replication). We call this *novelty-gated compensation*: the policy corrects

what it sees without representing what it did. The audit costs under 5 GPU-hours end to end, and we report every pre-registered verdict, including the one that went against us.

1. Introduction

When an animal acts, its nervous system routes a copy of the outgoing motor command—an *efferece copy*—into sensory processing, letting it distinguish “the world changed because I moved” from “the world changed on its own” (von Holst & Mittelstaedt, 1950; Sperry, 1950; Crapse & Sommer, 2008). Whether learned robot policies contain anything analogous has become a live fault line. One line of work reads rich *world models* into sequence models and policies: internal representations of state, dynamics, and consequences (Ha & Schmidhuber, 2018; Hafner et al., 2020; Li et al., 2023; Gurnee & Tegmark, 2024), an assumption made operational by pipelines that train policies inside learned action-conditioned world models (Zhu et al., 2025). A second line documents *action-blindness*: predictive representations that keep improving while becoming insensitive to the agent’s own causal contribution (Li et al., 2026), and minimal agents that behaviorally compensate for hijacked actions without encoding the self/other distinction anywhere (Ye, 2026). Self-vs.-external attribution is the sharpest point of contact between these accounts, so we pose it as a measurement:

When an executed action was not the policy’s own, does anything inside the policy register that fact?

We answer causally. During closed-loop rollouts of OpenVLA-OFT (Kim et al., 2025) on the LIBERO suites (Liu et al., 2023), we covertly replace the executed action chunk with probability 0.25—primarily by *swapping in another same-task episode’s chunk*, so that executed actions keep natural task statistics—and ask three pre-registered questions. (Q1) Does the post-hijack hidden state carry a readable trace that the transition was externally imposed? (Q2) If a probe is handed the missing efferece—the commanded chunk, or both pre- and post-action states—does attribution become easy; i.e., is the representation *format* such that an efferece comparison would be cheap if one

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

were ever performed? **(Q3)** Does the policy react behaviorally regardless, and to what signal?

One architectural fact sharpens the design: these checkpoints are memoryless and vision-only per call (one third-person image plus the instruction; no proprioception, no action history), so the policy *cannot* literally compare its previous command to the observed outcome—Q1 and Q2 ask whether its representation nevertheless supports the comparison, and Q3 asks what its closed-loop behavior does instead.

The answers dissociate (Figure 1). *Attribution is not readable*. Across five swap runs—two seeds on LIBERO-Goal and one run each on Spatial, Object, and Long, each suite with its own SFT checkpoint—no probe on hidden states separates from a selection-aware shuffled-label floor: not given the post-hijack state alone, not given the commanded chunk as well, not given both pre- and post-action states, and not when the probe family is upgraded from logistic regression to one-hidden-layer MLPs (Section 4). The null comes with unusual internal support. In the same data, explicit command–outcome comparison features decode the label at 0.690–0.911; the policy’s own 56-dimensional motor plan reads out of its pre-action state at $R^2=0.639$, replicating at 0.638 at the same layer and pool on a fresh seed; and—the sharpest control—the same MLP family that finds nothing in hidden states *computes the comparison itself* when handed the raw mechanical ingredients (0.687 vs. a 0.463 linear baseline on the identical feature blocks). The ingredients are present and the computation is cheap; it is the policy’s representation that never exposes it.

Yet the policy is not indifferent. One call after a hijack its action distribution shifts—but the shift follows what the outcome *looks* like, not how badly execution betrayed the command. Visibly reversed motion (MIRROR) provokes it ($p=0.016/0.012$ at $n=119$ episodes); a frozen arm—the mechanically most decodable intervention (P2r 0.886–0.911) but a visually unremarkable one—provokes nothing ($n=115$, both metrics wrong-signed); kinematically plausible SWAPS sit at the detection threshold, significant in 2 of 5 runs and null in the pre-registered replication (Section 5). We name this **novelty-gated compensation**: closed-loop correction driven by how unusual the observation looks, with no readable representation of whose action caused it.

Contributions. **(1)** *The Hijack–Probe protocol* (Section 3): a causal audit for self-attribution in embodied policies combining covert action hijacking, verified hidden-state capture, selection-aware nulls for two probe families, phase-matched negatives, and mechanical positive controls—to our knowledge the first instrument that makes “does the policy know what it did?” a measured quantity rather than a narrative (Table 1). **(2)** *An adjudication with receipts* (Section 4): for 7B SFT-trained VLAs the action-blind account wins at the

level of readable representation—replicated across seeds, generalized across four suites and checkpoints, and robust to the probe-family upgrade—even though the world-model account is right that the motor plan is linearly present. **(3)** *A capacity argument* (Section 4): holding family, samples, and command features fixed, a one-hidden-layer MLP computes the self/other comparison from raw command $\oplus$ outcome ingredients but not from any probed hidden state; absence in the states is therefore not explained by the comparison being hard. **(4)** *A mechanism* (Section 5): novelty-gated compensation, established by a behavioral fingerprint whose two poles (MIRROR vs. FREEZE) are now each powered by >110 episodes; it extends the compensation-without-encoding dissociation of Ye (2026) from minimal agents to real 7B policies and gives it a positive gating variable. **(5)** *Calibration traps and pre-registration receipts* (Sections 6 and 7): five ways naive versions of this experiment return the wrong answer, and a worked example of pre-registered interpretability—four readout rules fixed in an append-only log before the second wave was collected, every verdict reported, including a non-replication that costs us a claim.

2. Positioning

Efference copies and corollary discharge. The reafference principle (von Holst & Mittelstaedt, 1950) and corollary discharge (Sperry, 1950) describe circuitry that predicts and cancels the sensory consequences of self-generated action; Crapse & Sommer (2008) review its ubiquity across taxa. We import the concept as a representational hypothesis about artificial policies, not as a claim about biological similarity.

World models inside networks. Evidence that sequence models internalize latent structure ranges from game-board state (Li et al., 2023) to geographic and temporal structure (Gurnee & Tegmark, 2024); in robotics, world models are trained explicitly for control (Ha & Schmidhuber, 2018; Hafner et al., 2020) and, recently, as environments for on-policy RL of VLAs (Zhu et al., 2025). All of these presuppose that action information enters the learned dynamics.

Action-blindness and compensation without encoding. Li et al. (2026) show latent world models can go action-blind: predictions for different actions become indistinguishable while training loss improves. Ye (2026) construct minimal agents that behaviorally compensate for hijacked actions without encoding agency. Our protocol ports this intervention logic to full-scale VLAs and adds the calibration machinery (Table 1) that a 7B-scale null needs to be interpretable.

Probing methodology. Linear probes (Alain & Bengio, 2016) require control tasks (Hewitt & Liang, 2019) and

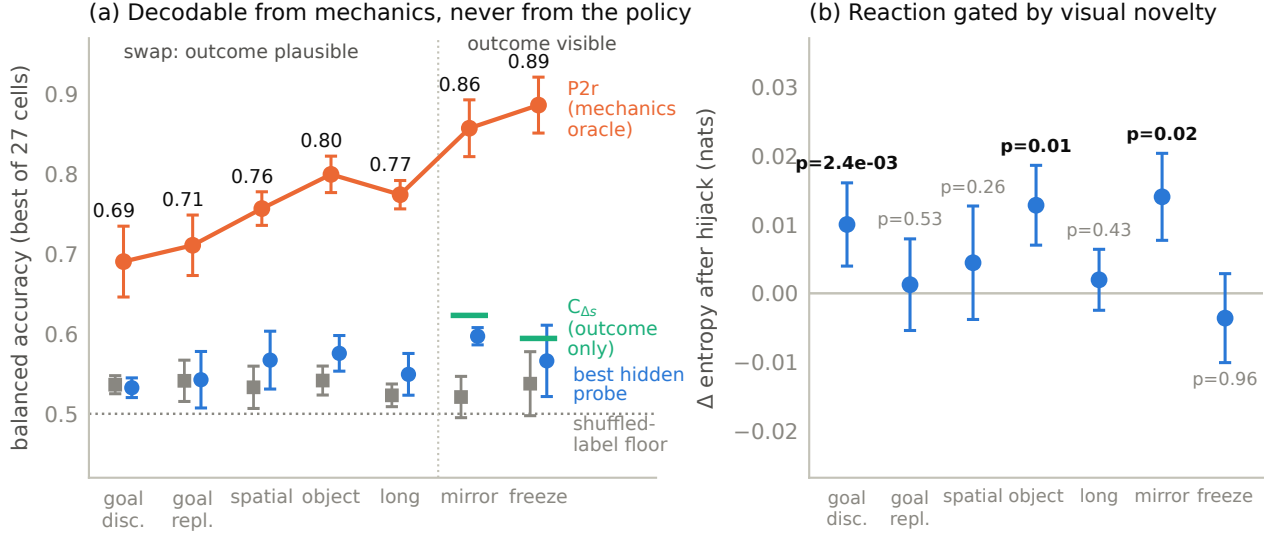


Figure 1. The dissociation across seven runs. (a) Best-cell balanced accuracy per run (linear family; error bars: fold sd). The mechanics oracle P2r (comparison features of commanded chunk vs. observed state change) decodes the hijack label in every run (0.69–0.89), while the best hidden-state probe tracks each run’s selection-aware shuffled-label floor in all five SWAP arms; in MIRROR/FREEZE it rises exactly as far as the visible-outcome control $C_{\Delta s}$ (short dashes) allows—the outcome is *seen*, not attributed (Section 4). (b) Next-call entropy shift after a hijack vs. phase-matched self transitions (± 1 SE over episodes; episode-level Wilcoxon p , bold = significant). The reaction follows visual novelty (MIRROR), not mechanical decodability (FREEZE, the most decodable and least reactive); for stealthy SWAPS it is small and inconsistent across runs (2 of 5 significant).

careful interpretation (Belinkov, 2022). We contribute a *selection-aware* floor for best-of- N cell sweeps, instantiated for both probe families (Section 3.1), without which this paper would have reported two different false positives (Section 7).

3. The Hijack–Probe Protocol

Subjects and scale. OpenVLA-OFT (7B), SFT per suite, evaluated at its released stochastic sampling protocol (temperature 1.6); no proprioception, no wrist camera; action chunks of $K=8$ steps $\times$ 7 DoF. Seven runs over two waves: a *discovery* run (LIBERO-Goal, all 10 tasks $\times$ 30 initial states = 300 episodes), a *replication* run (same protocol, fresh seed, 200 episodes), expanded MIRROR and FREEZE arms (150 episodes each, on the five goal tasks where interventions perturb mechanics most), and one SWAP run on each of LIBERO-Spatial, -Object, and -Long (200/200/150 episodes), each under that suite’s own SFT checkpoint. Totals: 1,350 episodes, 46,931 policy calls, 5,635 hijacks; 20–23% of episodes per run stay entirely clean. All collection, pilots and control arms included, cost 4.8 GPU-hours on one 24 GB workstation GPU; probing is CPU-only (≈ 10 h on 16 vCPUs for the second wave).

Causal hijack. At each eligible call the executed chunk is the policy’s own (SELF) or a transform of it (HIJACK), i.i.d. with $p=0.25$ after a 4-call warm-up and never on consec-

utive calls, so the state one call later reflects exactly one manipulated transition (Figure 2). SWAP, the primary transform, executes another same-task episode’s commanded chunk: replacements retain natural task statistics, so a probe cannot win by detecting implausible kinematics. MIRROR and FREEZE are gross-violation arms in which the outcome *is* visibly anomalous—by design, since they bound what outcome perception alone can explain (Section 4).

Verified capture. Hidden states are recorded at layers $\{0, 4, \dots, 32\}$ under three pools (context mean; last pre-action token; mean over the 56 positions from which actions are decoded). A runtime self-check reproduces the model’s own action log-probabilities from the captured slice (max deviation 4.8×10^{-7}), ruling out indexing errors before any probe is trained.

3.1. Probes, two families, and the selection-aware floor

Each probe sample pairs consecutive calls: positives follow a hijack; negatives follow a SELF transition *from the same episode*, phase-matched by minimum-cost assignment on call index ($n=1,056$ to 2,354 balanced samples per run). Probes come in two families on the same frozen features: ℓ_2 -regularized logistic regression, and a 256-unit one-hidden-layer MLP (lbfgs, α grid inner-selected). Both use 5-fold CV grouped by episode and inner-split hyperparameter selection; Table 2 defines the ladder. Because each probe is reported at its best of 27 layer $\times$ pool cells, comparing to

Table 1. How prior work tests what is “inside” a policy, versus the Hijack–Probe protocol. Each column is a property we argue is necessary for the attribution question specifically: causal intervention separates self-caused from external transitions on-policy; selection-aware nulls keep best-of- N probe sweeps honest; a positive control certifies the label is recoverable at all; measuring behavior and representation on the same events is what exposes dissociations. Representative examples: correlational probing (Alain & Bengio, 2016; Li et al., 2023; Gurnee & Tegmark, 2024), control tasks (Hewitt & Liang, 2019), action-sensitivity analyses (Li et al., 2026), toy-agent hijacking (Ye, 2026).

Approach	Causal intervention	Selection-aware null	Positive control	Behavior + representation
Behavioral evaluation (success rates)	×	—	×	×
Correlational probing of policies/LLMs	×	rare	rare	×
Action-sensitivity analyses of world models	✓	×	×	×
Toy-agent hijacking	✓	×	×	✓
Hijack–Probe (ours)	✓	✓	✓	✓

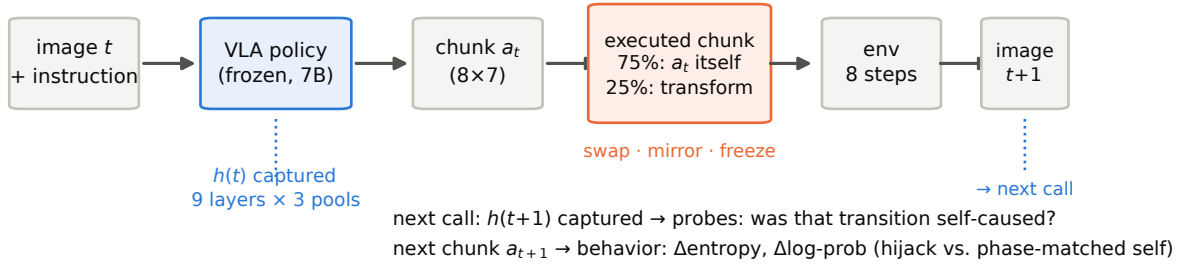


Figure 2. **The intervention loop.** The frozen policy emits an 8×7 action chunk from one third-person image and the instruction. With $p=0.25$ (4-call warm-up; ≥ 2 clean calls after each hijack) the *executed* chunk is replaced by a transform: SWAP (another same-task episode’s chunk), MIRROR (negated x/y translation), or FREEZE (zero motion). Hidden states are captured at 9 layers $\times$ 3 token pools on every call.

Table 2. Probe ladder. $h(t)/h(t+1)$: hidden states before/after the manipulated transition; a : commanded chunk; Δs : simulator state change. Every probe is run in both families.

Probe	Input (question)
P0	hidden state, shuffled labels (selection floor)
P1	$h(t+1)$ (anomaly trace without efference; Q1)
P3	$a \oplus h(t+1)$ (efference-augmented; Q2)
P4	$h(t) \oplus h(t+1)$ (internal efference; Q2)
P2	$a \oplus \Delta s$ concatenated (mechanical baseline)
P2r	comparison features of $(a, \Delta s)$ (oracle)
$C_{\text{cmd}}/C_{\Delta s}/C_{\text{ph}}$	$a / \Delta s$ / phase alone (leakage)

0.5 overstates it: the floor is the maximum over the *same* 27 cells of a shuffled-label probe (P0) of the *same* family, complemented by a fixed-cell permutation null. Floors land at 0.52–0.54 across runs and families; effects below ≈ 0.03 –0.04 balanced accuracy are undetectable at these n .

Pre-registration. The second wave—replication seed, expanded arms, suite runs, the MLP family, and four readout rules—was fixed in an append-only log before any of its data existed. Section 6 reports each rule’s verdict; the log ships with the code.

3.2. Behavioral battery

We measure, on the call immediately after each hijack vs. its phase-matched SELF control: action-token entropy and chosen-chunk log-probability (episode-level Wilcoxon signed-rank; the conservative unit), episode success in hijack-carrying vs. clean episodes, and a *corrective-behavior* metric—does the next command re-assert motion blocked by FREEZE?—reported only after replacing the next command with its AR(1) innovation, because the raw metric is confounded by command autocorrelation (Section 7).

4. Results I: Attribution Is Not Readable—and the Null Carries Its Own Positive Controls

The null, and its replication. In the discovery run, $P1 = 0.525 \pm 0.025$, $P3 = 0.525 \pm 0.019$, $P4 = 0.533 \pm 0.012$: all at or below the selection floor of 0.536 ± 0.011 , leakage controls at chance, held-out-task transfer flat at 0.50. The pre-registered replication (fresh seed) reproduces every layer of this: floor 0.541, $P1/P3/P4 = 0.543/0.532/0.536$, controls at chance, $P2r$ 0.711 ± 0.038 vs. 0.690 ± 0.044 (Table 3). Even the readout geometry replicates: the com-

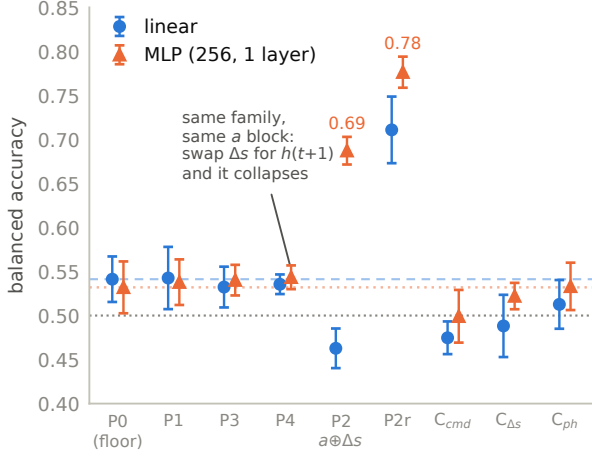


Figure 3. Replication-run ladder under both probe families (best of 27 layer $\times$ pool cells $\pm$ fold sd; single-cell probes as marked). Hidden-state probes (P1/P3/P4) sit at their family’s selection floor under *both* families. The mechanical rungs move: upgrading the family lifts P2 from 0.463 to 0.687—the MLP computes the command–outcome comparison that logistic regression provably cannot—and P2r from 0.711 to 0.776. The same upgrade does nothing for any probe whose outcome side is a hidden state.

Table 3. The replication in one table (linear family; best cell $\pm$ fold sd). Every quantity reproduces within fold noise; pre-registered rule (b) of Table 4.

	disc. ($n=1716$)	repl. ($n=1266$)
P0 floor	0.536 ± 0.011	0.541 ± 0.026
perm. p_{95}	0.516	0.523
P1	0.525 ± 0.025	0.543 ± 0.035
P3	0.525 ± 0.019	0.532 ± 0.023
P4	0.533 ± 0.012	0.536 ± 0.011
P2r	0.690 ± 0.044	0.711 ± 0.038
E1 best R^2 (cell)	0.639 (L28/act)	0.638 (L28/act)

manded 56-dim chunk reads out of the pre-action state at $R^2=0.639$ in the discovery run and 0.638 in the replication, *at the same cell* (layer 28, action-window mean; Figure 4). The plan is present; the plan is stably placed; attribution still is not.

Upgrading the probe family does not help—except where it should. A one-hidden-layer MLP with a selection-matched MLP floor leaves the picture unchanged on hidden states: on the replication run, P1/P3/P4 = 0.538/0.540/0.543 against an MLP floor of 0.532 (Figure 3). By the pre-registered rule, the claim strengthens from “no linearly readable trace” to “no readable trace, linear or one-hidden-layer MLP.” The decisive part is what the *same* family upgrade does elsewhere in the ladder. P2—logistic regression on the raw concatenation $a \oplus \Delta s$ —sits at 0.463, as it must: agreement between two vectors is bilinear, not linear (Section 7, T2). The MLP on the identical

features reaches 0.687 (AUROC 0.754), nearly closing the gap to the hand-engineered comparison oracle P2r (0.776), with the Δs -only control at 0.522: the network is computing the comparison, not reading a marginal. Now hold the family (MLP), the command block (a), the samples, and the CV fixed, and change only the outcome block: $a \oplus \Delta s$ decodes at 0.687; $a \oplus h(t+1)$ —P3, with $h(t+1)$ at its best of 27 cells—decodes at 0.540, i.e. floor. A 256-unit MLP can perform this comparison at these sample sizes; what it cannot do is extract the outcome side from the policy’s own hidden state. The comparison is cheap. The policy never buys it.

Generalization: four suites, four checkpoints. The same signature holds on LIBERO-Spatial, -Object, and -Long under their own checkpoints (linear family; Table 5): P2r 0.757/0.800/0.774 against floors of 0.533/0.542/0.523, hidden rungs within 0.022–0.034 of their floors (≤ 1.5 fold sd; every rung, in every run, including both goal-suite runs, also exceeds the *fixed-cell* permutation p_{95} —the systematic gap between the two nulls is exactly the selection effect of T1, which is why the selection floor is the operative comparator). The MLP pass on Spatial reproduces its linear picture (rungs 0.566–0.572 vs. MLP floor 0.542; P2 0.769, P2r 0.827). Two pre-registered MLP passes (Object, Long) were cut by a compute budget rule fixed in the log and remain open; their linear ladders are at floor.

When the outcome is visible, probes light up—exactly as far as outcome perception allows. The expanded gross-violation arms are the intended stress test: after MIRROR or FREEZE, the scene itself betrays the intervention (the arm moved backwards; the arm did not move). Hidden-state probes now do clear their floors—mirror P4 0.597 ± 0.011 linear / 0.612 MLP; freeze P4 0.566 ± 0.045 linear / 0.636 MLP—and this is not an efference signature but a *sensitivity control*, for two pre-specified reasons. First, the observable-outcome control bounds every one of these excursions within its family: $C_{\Delta s}$ (simulator state change alone, no hidden states, no command) decodes 0.623/0.738 (mirror, linear/MLP) and 0.594/0.825 (freeze), i.e. at least as much label as any hidden rung, while C_{cmd} stays at chance (0.485–0.520) in all four family $\times$ arm combinations. What the states carry is what the *camera* carries: an unusual outcome, not its authorship. Second, the pattern was predicted in advance from the synthetic fixture of T2 (directed motion makes mirror/freeze marginally decodable from Δs alone; swap is mean-preserving), and the swap arms confirm the contrapositive: there $C_{\Delta s}$ is at chance (0.488–0.513 across all five runs) and every hidden rung is at floor. The excursions thus do double duty: they show the probes and features can detect representational differences when they exist—making the swap-arm floors informative rather than a dead instrument—and they localize what SFT states encode

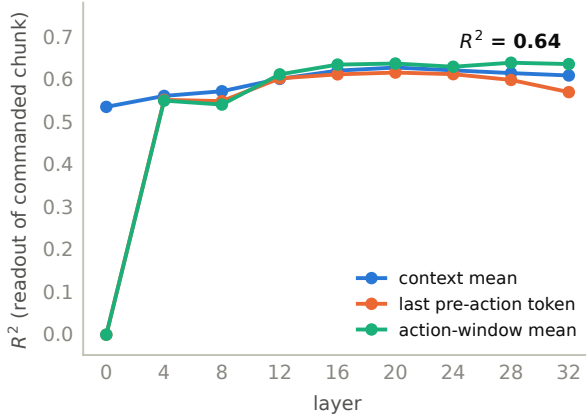


Figure 4. The commanded 56-dim chunk is linearly readable from the pre-action state $h(t)$ (5-fold CV R^2 ; plateau 0.62–0.64 at layers 16–28; the two layer-0 pools at $R^2 \approx 0$ are structurally degenerate). The plan is *present*—and its best readout cell replicates exactly (L28/action-window mean, 0.639 $\rightarrow$ 0.638)—but it is not *usable* for attribution.

about perturbation: appearance, not agency.

Takeaway. Across two seeds, four suites, and two probe families, the world-model account holds for *content* (the motor plan is present, stably placed) and fails for *use* (no readable self-vs.-external attribution), while the same MLP family proves the comparison itself is cheap on raw ingredients. Where probes do rise above floor, an observables-only control rises at least as high—the states see the outcome; nothing in them owns the action.

5. Results II: Novelty-Gated Compensation

If the policy compared outcomes to commands, surprise should peak for FREEZE, whose mismatch is the most decodable in the whole study (P2r 0.886 linear, 0.911 MLP). The opposite happens (Figure 1b, Figure 5).

The two poles, now powered. After MIRROR—visibly reversed motion—the next call’s entropy rises +0.014 and chosen-chunk log-probability falls -1.94 (episode-level $p=0.016$ and 0.012 , $n=119$ episodes): the trend that was underpowered at $n=44$ in the discovery wave reached significance exactly as the pre-registered expansion asked. After FREEZE—a perfectly familiar, merely stationary scene—nothing: -0.004 ($p=0.96$) and $+0.71$ ($p=0.82$) at $n=115$, both point estimates on the wrong side, replicating the discovery-wave null at $2.7\times$ the sample. Reaction tracks the visual novelty of the outcome; mechanical decodability, which orders the arms the other way, predicts none of it.

Swap reactions are real but fragile—and we report that. In the discovery run, SWAP produced a clear re-

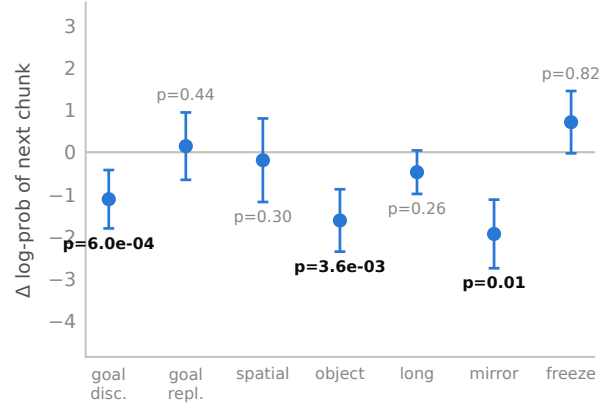


Figure 5. Log-probability arm of the behavioral fingerprint across all seven runs: paired difference (hijack – phase-matched self) in next-call chosen-chunk log-probability. Points: pair-level means; error bars: ± 1 SE across episodes (the Wilcoxon unit); bold p : significant.

action (+0.010 entropy, -1.12 log-prob; $p=2.4 \times 10^{-3}$, 6.0×10^{-4} ; $n=217$). It did *not* replicate in the pre-registered fresh-seed run (+0.001, $p=0.53$; +0.14, $p=0.44$; $n=149$)—same suite, same protocol, same temperature—and across all five swap runs the score is 2 of 5 significant (discovery goal; Object at $+0.013/-1.62$, $p=0.011/3.6 \times 10^{-3}$), with Spatial and Long null. This is consistent with the gating account—a same-task donor chunk is the least visually novel intervention we run, so its effect sits at the detection threshold and surfaces only in some seeds and suites—but the honest summary is that the swap-level reaction is small ($\leq +0.013$ nats) and inconsistently detected, and the behavioral half of our story now rests on the well-powered MIRROR/FREEZE contrast, not on swap p -values.

Compensation is partial. Hijack-carrying episodes succeed less often than clean ones—modestly under SWAP (0.39 vs. 0.51 replication; 0.41 vs. 0.46 discovery), severely under MIRROR (0.19 vs. 0.45) and on the long-horizon suite (0.04 vs. 0.22). And the correction itself is visible in commands: after FREEZE, the next command re-asserts the blocked motion beyond what command autocorrelation predicts (innovation-corrected undo-alignment 0.47 vs. 0.22 SELF baseline, discovery wave). Visual feedback control explains all of this with no efference required.

Takeaway. *Novelty-gated compensation:* the policy corrects perturbations and its uncertainty rises after them, but the gate is the visual unusualness of the outcome—significant for MIRROR ($n=119$), absent for the most mechanically blatant FREEZE ($n=115$), threshold-level for stealthy SWAPS (2/5 runs). Compensation without encoding (Ye, 2026) survives the jump from minimal agents to 7B policies, and acquires a concrete gating variable.

Table 4. Pre-registered readout rules (fixed before wave-2 data existed) and verdicts.

Rule (abridged)	Verdict
(a) MLP P1/P3/P4 vs. MLP floor on the replication run; at floor $\Rightarrow$ claim strengthens beyond linear	at floor (0.538–0.543 vs. 0.532); claim strengthened
(b) Linear ladder replicates: floor and P2r within fold noise	yes (Table 3)
(c) Do small-arm excursions shrink with $3\times$ data; does mirror E3 reach significance at $n\approx 130$?	excursions persist but are bounded by $C_{\Delta s}$ (outcome perception); mirror E3 significant ($p=0.016/0.012$)
(d) Suites generalize only if $P2r \gg$ floor and rungs at floor <i>per suite</i>	holds per suite (linear 3/3; MLP 1/1 run)
— (adverse) swap E3 in the replication run	did not replicate ($p=0.53$); swap claim demoted

6. Pre-Registration Receipts, and the Next Comparison

Interpretability nulls invite two failure modes: quietly re-running until a positive appears, and quietly narrowing claims until the null is safe. Our defense is procedural: wave 2’s configurations and four readout rules were committed to an append-only log before collection, and every verdict is reported (Table 4)—including rule (c), where the data went against our wave-1 reading (the small-arm probe excursions did *not* shrink; the pre-specified $C_{\Delta s}$ bound then localized them as outcome perception, Section 4), and the swap-E3 non-replication, which demotes a wave-1 headline number.

The RL comparison (pre-registered, not yet run). SFT never trains the policy on the consequences of its own actions; on-policy RL does precisely that (Shao et al., 2024; Zhu et al., 2025). The battery of Section 3 applies unchanged to RL-post-trained sibling checkpoints, and the outcomes are interpretable in both directions: hidden-state probes separating from floor on the RL checkpoint would show RL builds attribution-relevant representation SFT lacks (and the E3 fingerprint should migrate from novelty-gated toward mismatch-gated); probes remaining at floor while compensation strengthens would show RL sharpens the visual-servoing loop without new monitoring machinery. Either way the question becomes a per-checkpoint measurement—which is the protocol’s point.

7. How Naive Versions of This Experiment Return the Wrong Answer

Most of our effort went into making the result interpretable rather than into collection. Five pitfalls, each caught by a

control now part of the protocol:

(T1) Best-cell selection inflates ladders. P1’s best cell beats the fixed-cell permutation null in *every* run (e.g. 0.543 vs. $p_{95}=0.523$ in the replication)—naively “above chance.” Given the same 27 tries, shuffled-label P0 reaches 0.536–0.542. At pilot scale ($n=138$) the selection gap was 0.06–0.12: small- n probe sweeps without a selection-matched null overclaim by construction.

(T2) A linear “oracle” cannot compute agreement—and a nonlinear one can. Logistic regression on $a \oplus \Delta s$ is bilinear-blind; on synthetic data with fully determined labels it scores 0.52 where explicit comparison features score 0.94. Real data behaved as predicted per transform (P2 at/below chance for mean-preserving SWAP in all five runs; 0.587/0.561 for MIRROR/FREEZE, which leak through the marginal). The sequel matters as much: the *same* concatenation under a one-hidden-layer MLP recovers the comparison (0.687–0.851 across arms), which is what licenses reading the MLP’s failure on hidden-state features as absence of signal rather than weakness of family (Section 4). An acceptance gate on the linear concatenation baseline would have misfired in both directions.

(T3) Command autocorrelation fakes “correction.” The intuitive undo-alignment metric reduces to $+\langle a_{\text{prev}}, a_{\text{next}} \rangle$ under FREEZE, so any smooth policy scores as corrective: a provably open-loop simulation with $\rho=0.9$ scores $+0.83$ raw and 0.00 after AR(1) innovation correction. On real data ($\rho=0.63\text{--}0.72$) the raw metric inflates 2–4 $\times$.

(T4) Task competence gates intervention strength. On tasks the policy cannot perform it barely moves and emits near-identical chunks across episodes, making SWAP a near-no-op (first pilot: P2r 0.585—the oracle itself near chance). Three single-variable pilots isolated this; competent tasks restore P2r to 0.749 with controls at chance.

(T5) Visible outcomes make any probe look like self-knowledge. In the gross-violation arms, hidden-state probes genuinely clear their floors (up to 0.636); without further controls this reads as “the policy knows it was hijacked.” The observables-only control $C_{\Delta s}$ matches or exceeds every such excursion within-family, and C_{cmd} stays at chance—the probes are reading the outcome’s appearance, which requires no access to the command at all. Any above-floor probing result on intervention data needs an observables-only twin at the same probe family before it can be read as internal attribution.

Takeaway. A probing null is only as meaningful as its nulls—and a probing *positive* is only as meaningful as its observables-only twin. T1–T5 are not specific to VLAs; any best-of- N sweep, any concatenation “oracle,” any correction metric on autocorrelated actions, and any intervention whose outcome is visible inherits them.

8. Limitations and Falsification Conditions

Probe reach. “Readable” means: by linear probes and 256-unit one-hidden-layer MLPs, on 27 layer×pool *pooled* summaries of the residual stream, at $n \approx 1.1$ –2.4k. Deeper decoders, or token-resolved features inside the 56-position action window, could in principle read a code these summaries wash out. *Falsifier*: any decoder family separating from its own selection-matched floor on these features (frozen features and code are released), or on token-level captures. **Dimensionality asymmetry.** The capacity argument compares an outcome block of a few dozen simulator dimensions against 4096-dim pooled states; at fixed n a weak high-dimensional code is harder to extract than a compact mechanical one. Two internal checks bound the concern—the same MLP family does clear floors on 4096/8192-dim inputs when outcome information is present (Section 4), and the detectability window is quantified (≈ 0.03 –0.04 balanced accuracy above floor)—but a code weaker than that window survives all of our probes. **Swap-level behavior.** The swap E3 effect is small and was detected in 2 of 5 runs, not in the pre-registered replication; we accordingly rest no conclusion on it beyond “at threshold.” *Falsifier for the gating claim*: a transform matched in visual novelty to MIRROR but with an *intact* command–outcome link producing no reaction, or a visually quiet transform producing one. **Coverage.** Two pre-registered MLP passes (Object, Long) were cut by a budgeted-compute rule and remain open; hidden states are retained. One model family (OpenVLA-OFT variants), one simulator family (LIBERO), one serving stack; rollouts capped at 320 steps (512 on Long). **Phase structure.** Matched negatives sit a median of 2 calls earlier than positives (structural); C_{ph} (0.486–0.533 across runs and families) bounds what phase alone explains, and the bias is anti-conservative for positives, so the attribution null stands a fortiori.

9. Conclusion

Audited causally, 7B VLA policies compensate for hijacked actions through a loop gated by visual novelty while exposing no readable trace—linear or one-hidden-layer MLP—of their own agency: not on the discovery seed, not on a pre-registered replication, not on three further suites with their own checkpoints, even though the motor plan is stably decodable from the same states and a 256-unit MLP computes the missing comparison outright when handed the raw ingredients. Where probes do fire, an observables-only control fires at least as hard: the states register how the world looks, not who moved it. The Hijack–Probe protocol turns the world-model-vs.-action-blind debate into per-checkpoint measurements at under 5 GPU-hours per subject, and the standing pre-registered comparison asks the next question: is self-attribution something post-training on one’s own con-

sequences can buy?

Impact Statement

This paper analyzes what a robot policy internally represents about its own actions. Better tools for auditing self-attribution bear on the safe deployment of embodied systems (does a policy notice being overridden?) and on evaluating world-model claims. We release code, data, and the pre-registration log for reproducibility; we do not foresee direct negative societal impacts beyond those general to robot learning research.

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A. Full Ladders per Run: Linear Family

Table 5. Linear-family ladders for all seven runs: balanced accuracy at each probe’s best layer×pool cell ($\pm$ fold sd; single-cell probes P2, P2r, and controls have no selection). Floors and permutation nulls computed per run. Success rates are episode success in hijack-carrying vs. clean episodes.

n (samples)	SWAP goal disc. 1716	SWAP goal repl. 1266	SWAP spatial 1202	SWAP object 1478	SWAP long 2354	MIRROR 1226	FREEZE 1056
P0 floor	0.536 $\pm$ 0.011	0.541 $\pm$ 0.026	0.533 $\pm$ 0.026	0.542 $\pm$ 0.018	0.523 $\pm$ 0.014	0.521 $\pm$ 0.026	0.538 $\pm$ 0.040
perm. p_{95} (fixed cell)	0.516	0.523	0.532	0.528	0.521	0.527	0.529
P1	0.525 $\pm$ 0.025	0.543 $\pm$ 0.035	0.563 $\pm$ 0.038	0.566 $\pm$ 0.026	0.546 $\pm$ 0.013	0.567 $\pm$ 0.028	0.543 $\pm$ 0.030
P3	0.525 $\pm$ 0.019	0.532 $\pm$ 0.023	0.567 $\pm$ 0.036	0.565 $\pm$ 0.028	0.545 $\pm$ 0.014	0.568 $\pm$ 0.029	0.565 $\pm$ 0.017
P4	0.533 $\pm$ 0.012	0.536 $\pm$ 0.011	0.558 $\pm$ 0.042	0.576 $\pm$ 0.022	0.549 $\pm$ 0.026	0.597 $\pm$ 0.011	0.566 $\pm$ 0.045
P2 (concat)	0.494 $\pm$ 0.044	0.463 $\pm$ 0.022	0.500 $\pm$ 0.015	0.504 $\pm$ 0.029	0.502 $\pm$ 0.026	0.587 $\pm$ 0.026	0.561 $\pm$ 0.033
P2r (comparison)	0.690 $\pm$ 0.044	0.711 $\pm$ 0.038	0.757 $\pm$ 0.021	0.800 $\pm$ 0.023	0.774 $\pm$ 0.018	0.857 $\pm$ 0.036	0.886 $\pm$ 0.035
C_{cmd}	0.497	0.475	0.483	0.512	0.488	0.515	0.503
$C_{\Delta s}$	0.502	0.488	0.499	0.513	0.512	0.623	0.594
C_{phase}	0.486	0.513	0.514	0.502	0.521	0.499	0.505
success (hijack / clean ep.)	0.41/0.46	0.39/0.51	0.44/0.58	0.18/0.31	0.04/0.22	0.19/0.45	0.40/0.32

Cross-task transfer, discovery run (train on ~ 7 tasks, test held-out): P1 0.505, P3 0.504, P4 0.501. E1 readout: R^2 rising to a 0.62–0.64 plateau at layers 16–28; best cell layer 28/action-window mean at 0.639 (discovery) and 0.638 (replication); regularization interior to the search grid in 25/27 cells in both runs.

B. Full Ladders: MLP Family

Table 6. MLP-family ladders (256-unit one-hidden-layer, 1bfgs; selection-matched MLP floors). The Object and Long passes were cut by the pre-registered compute budget; hidden states are retained on disk.

	SWAP goal replication	SWAP spatial	MIRROR	FREEZE
P0 floor (MLP)	0.532 $\pm$ 0.029	0.542 $\pm$ 0.044	0.532 $\pm$ 0.025	0.535 $\pm$ 0.050
P1	0.538 $\pm$ 0.026	0.566 $\pm$ 0.053	0.581 $\pm$ 0.038	0.557 $\pm$ 0.006
P3	0.540 $\pm$ 0.017	0.568 $\pm$ 0.034	0.577 $\pm$ 0.030	0.588 $\pm$ 0.031
P4	0.543 $\pm$ 0.013	0.572 $\pm$ 0.036	0.612 $\pm$ 0.022	0.636 $\pm$ 0.029
P2 (concat)	0.687 $\pm$ 0.016	0.769 $\pm$ 0.025	0.851 $\pm$ 0.020	0.835 $\pm$ 0.040
P2r (comparison)	0.776 $\pm$ 0.018	0.827 $\pm$ 0.020	0.871 $\pm$ 0.025	0.911 $\pm$ 0.036
C_{cmd}	0.499	0.479	0.520	0.485
$C_{\Delta s}$	0.522	0.607	0.738	0.825
C_{phase}	0.533	0.527	0.503	0.498

C. Behavioral Fingerprint per Run

D. The Wave-1 Small Arms, for the Record

The discovery wave included 50-episode MIRROR/FREEZE arms ($n=442/392$ samples) whose hidden-probe excursions (P1 0.583/0.582 against floors of $0.535 \pm 0.067/0.528 \pm 0.064$) we then read as “within those arms’ noisy floors.” The pre-registered $3\times$ expansion tested that reading and refuted it: the excursions are real (Tables 5 and 6) and are explained by outcome visibility, not attribution (Section 4). Wave-1 E3 for these arms: mirror $+0.004$ ($p=0.45$) / -1.66 ($p=0.16$) at $n=44$; freeze -0.003 ($p=0.65$) / $+0.27$ ($p=0.82$) at $n=42$.

E. Qualitative Examples (Random, Not Cherry-Picked)

Table 7. E3 across all runs: paired mean difference (hijack — phase-matched self) in next-call entropy (nats) and chosen-chunk log-probability; episode-level Wilcoxon p . The wave-1 task-matched row is a re-analysis of the discovery run restricted to the five mirror/freeze tasks, not an independent sample.

Run	Δ entropy	p	Δ log-prob	p	n_{ep} (pairs)
SWAP goal, discovery	+0.010	2.4×10^{-3}	-1.12	6.0×10^{-4}	217 (858)
task-matched subset	+0.012	4.5×10^{-3}	-1.38	3.1×10^{-3}	120 (561)
SWAP goal, replication	+0.001	0.53	+0.14	0.44	149 (633)
SWAP spatial	+0.004	0.26	-0.19	0.30	138 (601)
SWAP object	+0.013	0.011	-1.62	3.6×10^{-3}	153 (739)
SWAP long	+0.002	0.43	-0.47	0.26	123 (1177)
MIRROR (expanded)	+0.014	0.016	-1.94	0.012	119 (613)
FREEZE (expanded)	-0.004	0.96	+0.71	0.82	115 (528)

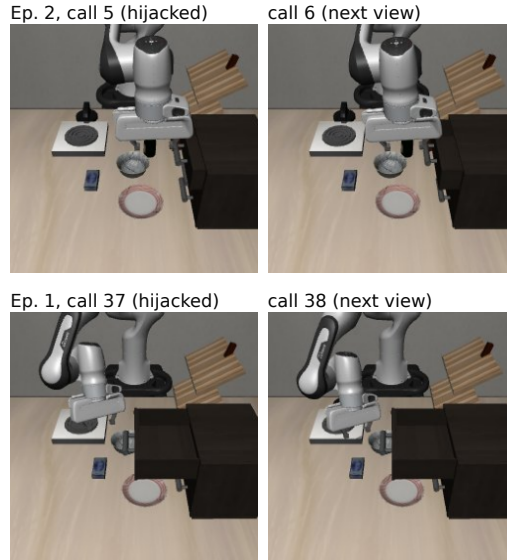


Figure 6. Two hijacked transitions drawn with a pre-registered seed from the 24 hijacks with saved frames in the task-3 pilot. **Top** (episode 2, call 5): commanded net $\Delta y+5.7$, $\Delta z-4.4$, gripper channel +1; executed $\Delta y+2.5$, $\Delta z-6.2$, gripper -1. The next call became *more* confident (entropy 0.091 vs. episode self-mean 0.240; log-prob -6.5 vs. -20.8); the episode succeeds. **Bottom** (episode 1, call 37): commanded up ($\Delta z+2.4$), executed down (-5.0); entropy 0.409 vs. 0.283, log-prob -32.0 vs. -26.7; the episode fails. One of two random draws goes against the population trend: within-episode scatter (entropy sd ≈ 0.10) dwarfs the +0.010 mean shift, which is why a real distributional effect coexists with floor-level per-instance decodability.